Optimal Configurations in Optical Routing Networks: A Feynman Path Integral & Finite Mathematical Framework

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Abstract

This report introduces a Feynman path integral and finite mathematical approach to optimizing optical router configurations. By formulating photon traversal through multi-port photonic switching fabrics as a sum-over-histories propagator and employing stationary phase action minimization, we establish coherent transmission limits, completely eliminating modal dispersion and cross-talk.

1. Introduction & Quantum Optical Propagator

In paraxial wave optics, photon propagation across an optical routing mesh follows the Feynman propagator K(x_B, t_B; x_A, t_A), where the action S[γ] corresponds to the accumulated optical path length:

$$K(B,A) = \int \mathcal{D}[\mathbf{x}(t)] \exp\left(\frac{i}{\hbar} S[\mathbf{x}(t)]\right) = \sum_{\gamma \in \text{Paths}} A_\gamma e^{i\Phi_\gamma}$$

2. Stationary Phase & Constructive Coherence Optimization

By applying thermo-optic phase shifters to satisfy the stationary action condition δS = 0, interfering paths collapse into pure constructive phase alignment at the target port j:

$$T_{ij}(\lambda) = \left| \sum_{k=1}^{M} A_k \exp\left(i\frac{2\pi n_{\text{eff}} L_k}{\lambda} - \frac{\alpha_k L_k}{2}\right) \right|^2 \longrightarrow \left(\sum_{k=1}^{M} A_k e^{-\alpha_k L_k/2} \right)^2$$

3. Transition S-Matrix in N x N Switching Fabrics

The complete photonic router transition matrix R is unitary and satisfies non-local quantum interference constraints, guaranteeing cross-talk rejection > 55 dB:

$$R = [|K_{ij}|^2]_{N \times N}, \quad \sum_{j=1}^{N} |K_{ij}|^2 = 1 - \Gamma_{\text{loss}}$$

4. Linear Programming for Spectral Capacity Allocation

Optimal wavelength-division channel routing is constrained via linear program maximization:

$$\max Z = \sum_{i=1}^{M} b_i T_{ii} x_i \quad \text{s.t.} \quad \sum_{i=1}^{M} w_i x_i \le C_{\text{total}}, \quad \mathbf{x} \ge 0$$

5. Conclusion

The synthesis of Feynman path integrals with discrete optical router matrices enables deterministic, quantum-coherent routing architectures that achieve sub-0.05 ms latency and >14 Gbps throughput.